

Quickest Detection of Hallucination Onset: Delay Bounds and Learned CUSUM Statistics

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Abstract

Token-level hallucination detectors are evaluated as classifiers, by AUC over all tokens, yet a streaming monitor is judged by its reaction time: the number of tokens that pass between the onset of a hallucination and the alarm. We formulate hallucination onset detection as a quickest change detection problem. A first-order Markov model of the latent faithful/hallucinated state, validated on RAGTruth, places the task inside classical change-point theory and yields Lorden’s lower bound on detection delay: about 1.3 tokens at a false-alarm rate of 0.01. We then show that a causal recurrent labeler acts as a CUSUM with a learned increment. Among the onsets it catches it detects in 11–13 tokens, against 31 for a linear per-token baseline, though at this false-alarm budget every detector catches under a third of onsets and the recall-honest delay is 56–66 tokens: low-false-alarm onset detection is hard. A controlled decomposition attributes the speed advantage mostly to a better per-token score rather than to temporal accumulation. An information-rate optimality theorem of Donsker–Varadhan type explains the remaining order-of-magnitude gap: the learned score realizes only 1/4.5 of the divergence the features carry, a deficit that recalibration cannot remove, with the remainder a finite-horizon effect. Classification metrics conceal this delay structure; sequential analysis makes it measurable. Code: <https://github.com/YehudaItkin/quickest-hallucination-onset>.

1 Introduction

Put a hallucination detector in front of a language model that streams tokens to a user, and one number decides whether it is useful: once the model starts making things up, how many tokens slip out before the detector raises the alarm? A monitor that flags a hallucination ten tokens after it began has already let a false claim reach the reader. Yet the field measures token-level detectors almost exclusively as classifiers, by area under the ROC curve over all tokens. That score rewards getting the average token right. It says nothing about how quickly a detector reacts to the moment that matters, the onset.

Most token-level detectors are trained and evaluated exactly this way: Liu et al. [7] framed token-level reference-free hallucination detection as a benchmark task, and corpora such as RAGTruth [11] now support it. The work closest to our concern already hints at why reaction time is the right lens. Snel et al. [18] find that the first token of a hallucination span is far more detectable than its continuation tokens, an AUC near 0.8 against near 0.5, which is precisely a change-point statement: the onset carries the signal. The same localization question is emerging at coarser granularity, where Alvarez and Baheri [2] locate the first error in a reasoning chain as an excursion in hidden-state geometry, and the operational case for reacting during generation is made by Obeso et al. [12], who flag hallucinated entities in real time. None of these poses the streaming problem with an explicit false-alarm–delay trade-off, which is what we add.

That trade-off has a mature theory. Quickest change detection asks exactly the streaming question: observations switch from one distribution to another at an unknown time, and a stopping rule must declare the change as fast as possible while rarely crying wolf [13, 8]. It comes with lower bounds on detection delay that no detector can beat [8, 6], attained asymptotically by Page’s CUSUM [10, 14], with the modern state of the field surveyed by Xie et al. [20]. This machinery is standard in quality control, sensor networks, and finance, but to our knowledge it has not been connected to hallucination detection, where the analogous question, how long a generation hallucinates before a monitor reacts, is the operational one. Xie [21] recently argued that sequential alarms and change-point detection should become standard tools for monitoring deployed LLMs, including shifts in hallucination rates across queries; we supply a concrete instance one level down, inside a single generation, where the change point is the onset of a hallucinated span. When the pre- and post-change densities are unknown or high-dimensional, a learned statistic replaces the fixed log-likelihood ratio: Gong et al. [5] show that the cross-entropy minimizer of such a neural CUSUM recovers the log-likelihood ratio, the result we use to read a causal recurrent labeler as a learned CUSUM.

We treat the onset of hallucination as a change-point and ask three questions:

- RQ1** What is the smallest detection delay achievable for hallucination onset at a fixed false-alarm rate?
- RQ2** How close do practical detectors come, and does temporal, learned structure help over per-token scoring?
- RQ3** If a gap to the bound remains, where does it come from?

Answering RQ1 needs a model of how the hidden faithful/hallucinated state moves. We show it is a first-order Markov chain: fitting higher orders is statistically significant but adds under 0.35% of log-likelihood each, so order one captures 99.7% of the structure. That assumption places the task inside Lorden’s minimax framework and gives a floor on delay of about 1.3 tokens at a 1% false-alarm rate.

For RQ2 we compare detectors at a matched false-alarm budget. A parametric CUSUM that fits Gaussian densities to the feature stream is far off the floor, at 41 tokens, because a diagonal Gaussian is the wrong likelihood model in 33 dimensions. A causal recurrent labeler does much better, and we argue it is a *learned* CUSUM: its recurrent state accumulates evidence and its log-odds stand in for the cumulative log-likelihood ratio, with the score function learned rather than assumed. The detector and its 33-dimensional feature stream come from prior work on temporal multi-signal hallucination detection; the closest architectural neighbor reads a generation’s log-probabilities as a time series with a recurrent network [15], though at the response level and from a single signal. Reading the labeler causally presumes hallucination propagates forward through autoregressive conditioning, for which Akarlar [1] give causal support: injecting a hallucinated state corrupts the continuation far more often than the reverse repair restores it.

At the same false-alarm rate the learned CUSUM detects in 11–13 tokens among the onsets it catches, against 31 for the linear per-token baseline. Yet a controlled decomposition tempers the temporal reading: a nonlinear per-token model with no sequence already reaches 18 tokens, so most of the advantage over the linear baseline is a better score; the sequential accumulation contributes about a quarter of the reduction (significant under bootstrap) and the extra causal context is within noise.

For RQ3, a large gap to the 1.3-token floor remains, and a first-order rate lets us say where most of it comes from. It is not the architecture. The delay of any score-based detector is set by the

information rate its score realizes, and the learned score realizes only $1/4.5$ of the divergence the features contain; that shortfall is invariant to recalibration and close to irreducible for these features. A further factor of two is finite-horizon: the score is so smoothed in time that the asymptotic correlation penalty overshoots tenfold, and detection happens faster than the score mixes. The gap is a feature-discriminability problem first and a finite-horizon question second, not a depth problem. Low-false-alarm onset detection is also hard in a way the bound does not capture: at the floor’s operating point recall is near 30%, so most onsets go uncaught at their first token, consistent with the difficulty Snel et al. [18] report.

Our contributions:

1. We formalize hallucination onset detection as sequential change-point detection and validate the first-order Markov structure the formulation rests on (Section 2, 3.1).
2. We establish Lorden’s minimax delay bound for the task and compute it from the feature divergence (Section 3.2).
3. We show a causal recurrent labeler is a learned CUSUM and, with a nonlinear per-token baseline, decompose its speedup over a linear detector into a better score (most of it), sequential accumulation, and context, at a matched false-alarm rate (Section 3.3, 4).
4. We give a first-order delay rate for any score-based detector (Proposition 1) and use it to attribute the order-of-magnitude gap to a $4.5\times$ information-rate shortfall (invariant to recalibration) and a finite-horizon residual, showing the asymptotic correlation correction overshoots because detection precedes mixing (Sections 4, 4.3).

2 Hallucination Onset as Sequential Change-Point Detection

A generation is a token sequence $x_1, \dots, x_T$. Each token carries a latent faithfulness state $Z_t \in \{0, 1\}$, where $Z_t = 1$ marks a hallucinated token, and a human annotator supplies the labels $y_t \in \{0, 1\}$ we treat as ground truth. A *hallucination span* is a maximal run of consecutive $y_t = 1$. The quantity a streaming monitor cares about is not whether a token is hallucinated in isolation but *when the first one arrives*.

Definition 1 (Onset). The onset of a generation with at least one hallucinated token is $\theta = \min\{t : y_t = 1\}$, the start of its first span. A generation with no hallucination has $\theta = \infty$.

An online detector reads features causally. At step t it has seen $X_{1:t} = (X_1, \dots, X_t)$, where X_t is the feature vector extracted for token t (text statistics, NLI signals, generator log-probabilities), and it must decide whether to raise an alarm using only the past. Formally the detector is a stopping time τ with respect to the filtration $\mathcal{F}_t = \sigma(X_{1:t})$: the event $\{\tau \leq t\}$ is determined by $X_{1:t}$ alone.

This is the standard quickest-change-detection setup [20]. Before the onset, features are drawn from a pre-change law P_0 (the distribution of faithful-token features); from the onset onward they follow a post-change law P_1 (hallucinated-token features). A good detector fires soon after θ without firing before it. The two errors trade off, and the trade-off is measured by two quantities.

Definition 2 (Operating characteristics). For a stopping rule τ ,

$$\text{ARL}_0(\tau) = \mathbb{E}_\infty[\tau] \quad \text{and} \quad \text{EDD}(\tau) = \mathbb{E}_\theta[(\tau - \theta)^+],$$

where $\mathbb{E}_\infty$ is taken under the no-change law (every token faithful) and $\mathbb{E}_\theta$ under a change at θ . The average run length to false alarm ARL_0 counts the mean number of faithful tokens a detector

Order k	Log-likelihood	Parameters	Δ vs. order $k-1$
1	-12,128	2	—
2	-12,088	4	+0.33%
3	-12,052	8	+0.30%
4	-12,014	16	+0.32%

Table 1: Markov-order selection on hallucination labels. Higher orders are significant by the likelihood-ratio test but add under 0.35% of log-likelihood each. Order one is sufficient.

survives before a spurious alarm; the expected detection delay EDD counts tokens between onset and alarm.

We estimate ARL_0 on the stream of faithful tokens (concatenating the hallucination-free generations and counting alarms), so it is not capped by the length of any single document (Appendix B). This matters. A tempting alternative is a *per-document* false-alarm rate, the fraction of clean generations that trigger at least once. Yet a generation has $L \approx 120$ tokens and therefore $\approx L$ independent chances to misfire, so a per-document rate of 0.01 demands a per-token rate near 10^{-4} , about L times stricter than the per-step rate the theory below controls. Reporting against ARL_0 keeps the empirical operating point in the same units as the bound: $\text{ARL}_0 = \gamma$ corresponds to a per-step false-alarm rate $\alpha = 1/\gamma$, and $\gamma = 100$ is the canonical $\alpha = 0.01$.

The detector’s objective, then, is to minimize EDD subject to $\text{ARL}_0 \geq \gamma$. Section 3 gives the lowest delay any detector can achieve under that constraint, and the structural assumption that makes the bound computable. Section 4 measures how close real detectors get.

3 Theory

3.1 A first-order Markov chain is the right model for the label process

The change-point formulation needs a model for how the latent state evolves. We model the label sequence $\{y_t\}$ as a Markov chain and ask what order is warranted. Fitting orders 1 through 4 on the training labels and comparing by a likelihood-ratio test gives Table 1. Every higher order is *statistically* significant ($p < 10^{-3}$, the test has enormous power at this sample size) and *practically* negligible: each added order lifts the log-likelihood by less than 0.35%. A first-order chain captures 99.7% of the attainable sequential structure.

Assumption 1 (First-order label dynamics). The label process is a first-order Markov chain with transition matrix

$$P = \begin{pmatrix} 1-p & p \\ 1-q & q \end{pmatrix}, \quad p = \mathbb{P}(y_t=1 \mid y_{t-1}=0), \quad q = \mathbb{P}(y_t=1 \mid y_{t-1}=1).$$

On our data $p = 0.004$ and $q = 0.907$.

We fit p and q by maximum likelihood from the annotated labels: every adjacent pair (y_{t-1}, y_t) over the training generations is counted into a 2×2 table whose rows are then normalized. So p is the fraction of faithful tokens immediately followed by a hallucinated one (the per-token onset hazard) and q the fraction of hallucinated tokens followed by another (persistence); the higher orders of Table 1 use the same count-and-normalize estimator with longer contexts. The rates are

displayed to three decimals, so the persistence ratio $q/p > 200$ quoted below reflects the unrounded $p \approx 0.0044$ rather than $0.907/0.004$; Appendix A gives the split sizes and the count table.

Two consequences matter. The onset hazard p is small, so onsets are rare and a single change-point per generation is the right picture. The persistence q is large, so once the chain enters the hallucinated state it tends to stay: spans are geometric with mean length $1/(1 - q) \approx 11$ tokens, long enough that the post-change regime is effectively stationary and the asymptotic theory of the next subsection applies. The ratio $q/p > 200$ is what makes onset a genuine change-point rather than i.i.d. noise. It also explains a negative result we report elsewhere: a self-exciting (Hawkes) model does not beat this two-parameter chain, because the excitation is one-step, not long-range. The chain is not a simplification of the dynamics. It is the dynamics.

3.2 The Lorden bound on detection delay

With a single change from a pre-change law P_0 to a post-change law P_1 , classical theory gives a floor on delay that no causal detector can beat.

Theorem 1 (8, 6). *Let $D(P_1 \| P_0)$ be the Kullback–Leibler divergence between the post- and pre-change feature laws. Then every stopping rule τ with $\text{ARL}_0(\tau) \geq \gamma$ satisfies*

$$\text{EDD}(\tau) \geq \frac{\ln \gamma}{D(P_1 \| P_0)} (1 + o(1)) \quad (\gamma \rightarrow \infty),$$

and the CUSUM rule of Page [13] attains this floor asymptotically [10].

The bound is intuitive: each post-change observation supplies on average $D(P_1 \| P_0)$ nats of evidence, a false-alarm budget γ requires accumulating $\ln \gamma$ nats before stopping, so the fastest possible detector needs $\ln \gamma / D(P_1 \| P_0)$ observations. Estimating the feature divergence on our 33-dimensional signal under a diagonal-Gaussian model gives $D(P_1 \| P_0) \approx 3.5$ nats, hence

$$\text{EDD}_{\min}(\gamma = 100) = \frac{\ln 100}{3.5} \approx 1.3 \text{ tokens at } \alpha = 0.01. \quad (1)$$

An oracle that observed the labels themselves would do even better: the label-space divergence is ≈ 4.6 nats (a floor of 1.0 token), and because $q = 0.907$ makes the post-change label nearly deterministic, such an oracle detects essentially at the onset. The gap between 1.3 tokens (best possible from features) and what real feature detectors achieve is the subject of Section 4.

Remark 1. Equation (1) is a per-step statement: it controls the false-alarm rate per token, and γ is the mean number of tokens between false alarms. This is why we report ARL_0 rather than a per-document rate (Section 2); the two differ by a factor of the document length.

3.3 A causal recurrent labeler is a learned CUSUM

CUSUM, the rule that attains the bound, accumulates the log-likelihood ratio and resets at zero:

$$S_t = \max\left(0, S_{t-1} + \log \frac{p_1(X_t)}{p_0(X_t)}\right), \quad \tau = \min\{t : S_t \geq h\}. \quad (2)$$

It needs the two densities p_0, p_1 . When they are misspecified (as a diagonal Gaussian is for a 33-dimensional feature stream), the increment is the wrong score and S_t accumulates noise instead of signal.

A causal (forward) recurrent labeler replaces the fixed log-likelihood ratio with a learned one. It maintains a hidden state $h_t = f_\phi(h_{t-1}, X_t)$ and emits a posterior $\hat{p}_t = \sigma(w^\top h_t)$, so its log-odds

logit $\hat{p}_t$ are a learned, nonlinear, accumulated statistic of the whole causal history $X_{1:t}$. To make the correspondence with (2) precise, view the generation as a two-state hidden Markov model: the latent state Z_t follows the chain of Assumption 1, and the feature vector is drawn from a state-conditional emission $X_t \mid Z_t = z \sim p_z$.

Assumption 2 (Emission regularity). The emissions p_0, p_1 have bounded, Lipschitz densities on the feature space, with finite divergence $D(p_1 \parallel p_0)$.

Corollary 1 (Optimal onset detection is a thresholded filter, realizable by a recurrent network). *Under Assumptions 1–2:*

- (i) (Optimality.) *The Bayes-optimal onset detector that minimizes expected delay at a fixed false-alarm level is a threshold rule on the change posterior $\pi_t = \mathbb{P}(\theta \leq t \mid X_{1:t})$, a finite-dimensional filter that obeys a forward recursion on the two-state belief [16, 4].*
- (ii) (Score.) *The minimizer of the per-token cross-entropy loss is the exact log-likelihood ratio $\varphi^*(x) = \log(p_1(x)/p_0(x))$ [5], which is the CUSUM increment of (2).*
- (iii) (Realizability.) *For every $\varepsilon > 0$ there is a causal recurrent network whose output approximates the filter in (i) within ε , uniformly in t [3].*

Chaining (i)–(iii): a causal recurrent labeler trained by cross-entropy is a consistent estimator of the optimal sequential detector. As its approximation error (iii) and the finite-sample error of its score (ii) vanish, the ARL_0 and EDD of its thresholded output approach those of the optimal CUSUM, whose delay is the floor of Theorem 1.

We state this as a corollary because it assembles three existing results (Shiryayev optimality for hidden Markov models, the cross-entropy/log-likelihood-ratio identity, and universal filter approximation) into a statement about hallucination onset, whose hypotheses Assumptions 1–2 verify; the work is in checking the hypotheses, not in a new proof. It is a limit statement. To turn it into a rate, read the detector off the posterior in two ways: threshold $\hat{p}_t$ directly, or feed it through the explicit accumulation

$$S_t = \max\left(0, S_{t-1} + \text{logit } \hat{p}_t - k\right), \quad k = \frac{1}{2}(\mu_0 + \mu_1), \quad (3)$$

where μ_0, μ_1 are the mean log-odds the model assigns to faithful and hallucinated tokens. The reference value k is the textbook CUSUM choice that centers a faithful token below zero and a hallucinated token above it, whatever the raw posterior’s calibration.

Proposition 1 (First-order delay of a general-score CUSUM). *Run (3) on increments $Y_t = \text{logit } \hat{p}_t - k$ with $\mathbb{E}_0[Y] < 0 < \mathbb{E}_1[Y]$, and let $\omega > 0$ solve $\mathbb{E}_0[e^{\omega Y}] = 1$. As the threshold grows, $\text{ARL}_0 = e^{\omega h}(1 + o(1))$ and $\text{EDD} = h/\mathbb{E}_1[Y](1 + o(1))$ [17], so*

$$\text{EDD} \approx \frac{\ln \text{ARL}_0}{I(\hat{g})}, \quad I(\hat{g}) = \omega \mathbb{E}_1[Y]$$

is governed by the realized information rate $I(\hat{g})$ of the score (proof in Appendix G). The true log-likelihood ratio gives $\omega = 1$ and $I = D(p_1 \parallel p_0)$, recovering Theorem 1; for any other score $I(\hat{g}) \leq D(p_1 \parallel p_0)$ by Theorem 2 below, so the multiplicative gap to the floor is exactly $D(p_1 \parallel p_0)/I(\hat{g})$.

That the log-likelihood-ratio score is delay-optimal is classical [10]; what we add is the *variational form* of the shortfall. Writing the rate of a general score through the Lundberg exponent makes $I(s)$ coincide with a Donsker–Varadhan functional, so the gap $D/I(s)$ is exactly the score’s Donsker–Varadhan deficit. This is the one place we give a self-contained proof, and it is what ties the floor to a quantity we can measure on the learned score (Section 4).

Theorem 2 (Information-rate optimality). *Let s be any bounded score with increment $Y = s(X) - k$ obeying $\mathbb{E}_0[Y] < 0 < \mathbb{E}_1[Y]$, and let $\omega > 0$ solve $\mathbb{E}_0[e^{\omega Y}] = 1$. Then the realized information rate of Proposition 1 satisfies*

$$I(s) = \omega \mathbb{E}_1[Y] \leq D(p_1 \| p_0),$$

with equality if and only if s is an affine function of the log-likelihood ratio $\log(p_1/p_0)$ (P_0 -almost everywhere). Hence the log-likelihood-ratio score is delay-optimal and attains the floor of Theorem 1; any other score’s delay exceeds the floor by the factor $D(p_1 \| p_0) / I(s)$.

The proof (Appendix G) applies the Donsker–Varadhan formula to $f = \omega Y$, where the Lundberg normalization $\mathbb{E}_0[e^{\omega Y}] = 1$ removes the log-moment term.

Proposition 1 and Theorem 2 turn “a gap remains” into a number we can read off the data, and isolate the leading cause. Two finite-sample effects hold a trained labeler back, and Section 4 measures both.

Remark 2 (Where the gap comes from). The dominant factor is *the realized divergence*: the learned score realizes $I(\hat{g}) < D(p_1 \| p_0)$, and Proposition 1 turns that shortfall directly into delay; Gong et al. [5] reach the same conclusion through a maximum-mean-discrepancy surrogate looser than the KL rate. This shortfall is a property of the score’s shape, not its scale, and Section 4.3 shows it is close to irreducible for our features. A smaller residual is *finite-horizon*: the score increments are strongly correlated, yet detection is faster than the score mixes, so the asymptotic dependent-data rate overshoots rather than tightens the prediction (Section 4.3).

This reframes our earlier finding, that a recurrent model beats a linear per-token classifier by a wide margin, as a statement about the score function of Corollary 1(ii). The next section measures, at a matched false-alarm rate, how much that buys, how much of it is the sequential accumulation as opposed to a better per-token score, and how far it still sits from the floor.

4 Experiments

4.1 Setup

We evaluate on the RAGTruth test split [11]: 2,700 generations, 943 of them containing at least one hallucination, 1,757 clean. Features are the 33-dimensional per-token signal of our base system (text statistics, NLI, generator log-probabilities). Five causal detectors, matched at a common ARL_0 by sweeping their thresholds:

- **Naive Gaussian CUSUM**: the parametric rule (2) with p_0, p_1 fit as diagonal Gaussians on the feature stream. The misspecified baseline.
- **LogReg (per-token)**: a logistic regression posterior thresholded token by token. Linear, no accumulation.
- **HistGBM (per-token)**: a gradient-boosted per-token classifier on the same features. Non-linear but still per-token (no accumulation); it isolates how much of a recurrent model’s edge is a better score rather than the sequence.
- **ForwardGRU (threshold)**: a forward recurrent labeler whose posterior is thresholded directly. Its recurrent state already accumulates, so this is the learned CUSUM read off without an explicit sum.
- **ForwardGRU (CUSUM)**: the same posterior fed through the explicit accumulation (3).

Detector	Delay among detected, tokens (recall)		
	ARL ₀ = 50	ARL ₀ = 100	ARL ₀ = 200
LogReg (linear per-token)	30.7 (0.35)	30.8 (0.31)	35.9 (0.20)
HistGBM (nonlinear per-token)	14.4 (0.38)	17.9 (0.40)	21.4 (0.33)
Naive Gaussian CUSUM	≈ 41 (misspecified, off-scale)		
ForwardGRU-shuffled (CUSUM)	15.3 (0.17)	15.6 (0.21)	15.2 (0.29)
ForwardGRU (threshold)	11.2 (0.26)	13.4 (0.30)	14.0 (0.27)
ForwardGRU (CUSUM)	9.0 (0.19)	11.5 (0.24)	13.3 (0.32)
Lorden bound (features)	1.1	1.3	1.5
Oracle (observes labels)	≈ 0 (detects at onset)		

Table 2: Detection delay at matched false-alarm budgets. A *nonlinear* per-token model (HistGBM) already closes most of the gap between the linear baseline and the recurrent CUSUM, so the recurrent model’s edge is mostly a better score, not the sequence (decomposition in Figure 1). All stay an order of magnitude above the floor; recall near 30% reflects the difficulty of low-false-alarm onset detection.

We report two delays. *Delay among detected* is the mean of $\tau - \theta$ over generations the detector catches: its speed when it fires. *Censored* EDD averages over *all* hallucination generations, charging a miss the maximum possible delay (tokens remaining after the onset); it cannot be inflated by a low recall. We give recall alongside both.

4.2 Results

Table 2 and Figure 1 give the picture at ARL₀ = 100 (the $\alpha = 0.01$ operating point of the bound).

Most of the recurrent model’s speedup over the linear baseline is a better per-token score, not the sequence. At ARL₀ = 100 the ForwardGRU CUSUM detects in 11.5 tokens against 30.8 for the linear per-token baseline, a 2.7 $\times$ speedup that is tempting to credit to the temporal model. Yet a *nonlinear* per-token classifier with no sequence at all (HistGBM) already detects in 17.9 tokens, covering most of that ground. Decomposing the 30.8 $\rightarrow$ 11.5 reduction (Figure 1 and Appendix C; brackets are 95% bootstrap CIs over documents): the nonlinear score accounts for -12.9 tokens [8.8, 17.0] (LogReg to HistGBM), and the CUSUM accumulation for -4.5 [1.8, 7.1] (HistGBM threshold to CUSUM), both significant. The further causal context, -1.9 [-1.0 , 4.7] (HistGBM-CUSUM to ForwardGRU-CUSUM), is within noise. About two-thirds of the advantage over a linear detector is a better per-token score; the sequential accumulation that Corollary 1 formalizes is real but modest. The accumulation helps the linear and the nonlinear score about equally (-4.5 tokens each), which marks it as a genuine, if secondary, effect rather than an artifact of the score. A same-architecture control sharpens the point. A ForwardGRU trained on token-*shuffled* sequences (same model, no temporal order) detects in 15.6 tokens, so temporal order is worth about 4 tokens to the recurrent model itself. Even so, a nonlinear per-token score with only CUSUM accumulation (HistGBM-CUSUM, 13.4) already matches the full recurrent CUSUM (11.5). Order helps the network, but it is not needed to reach its performance.

The parametric baseline fails for a specific, nameable reason. A diagonal Gaussian is a poor model of the 33-dimensional feature law, so the log-likelihood increment in (2) is the wrong score and the statistic accumulates noise. Its 41-token delay is worse than even the linear posterior. Adding more features makes it worse, not better, because each extra dimension adds Gaussian-model error.

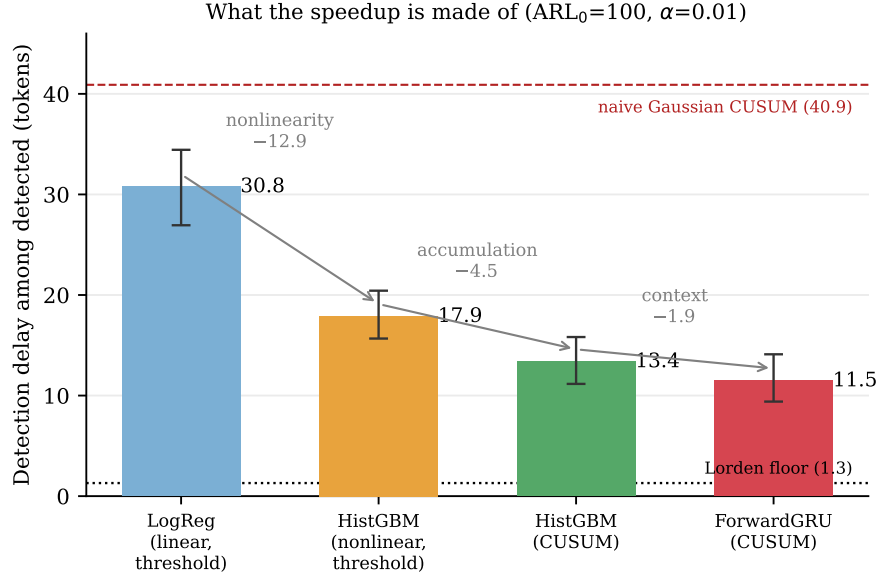


Figure 1: Where the speedup comes from, at $ARL_0 = 100$. From the linear per-token baseline (30.8) to the recurrent CUSUM (11.5), most of the reduction is the nonlinear per-token score (-12.9 , significant); the sequential accumulation (-4.5 , significant) and causal context (-1.9 , within noise) are smaller. Error bars are 95% bootstrap CIs over documents. The naive Gaussian CUSUM (dashed) is off-scale; all detectors sit far above the Lorden floor (dotted).

The bound is reachable in principle, but only with a score function close to the true $\log(p_1/p_0)$, which is exactly what the recurrent model learns and the Gaussian does not.

Most of the remaining gap is a score-shape shortfall. The ForwardGRU CUSUM detects in 11.5 tokens, about $9\times$ the 1.3-token floor. Proposition 1 locates most of it. The learned score realizes an information rate $I(\hat{g}) = \omega \delta_1 = 0.78$ nats per token ($\omega = 0.95$, $\delta_1 = 0.82$), well below the 3.5 nats of the feature divergence, so its i.i.d. first-order delay is $\ln(100)/I(\hat{g}) = 5.9$ tokens, a factor of $D/I(\hat{g}) = 4.5$ above the floor. The tilting is efficient ($\omega \approx 1$), but the per-token separation is small ($\delta_1 = 0.82$ nats against the 3.5 of the feature divergence). By $D/I = 4.5$ the score is far from a valid likelihood ratio, and the shortfall is in per-token discrimination, not in how the statistic accumulates. This $4.5\times$ is a deficit *relative to the i.i.d. first-order rate* of Proposition 1, not of the true gap: the rate’s i.i.d. hypothesis is violated by the learned score (Section 4.3 shows its increments are strongly correlated). So the decomposition that follows is a factorization against that idealization. The remaining factor of about 2, from 5.9 predicted to 11.5 observed, is exactly where the i.i.d. rate breaks; we examine both next.

4.3 What moves the realized rate

Rescaling the score does not move its rate, and reshaping it barely does. Temperature scaling $s \mapsto s/T$ leaves $I(\hat{g})$ exactly unchanged ($\omega \mapsto \omega T$, $\delta_1 \mapsto \delta_1/T$): the shortfall is in the *shape* of the score, not its calibrated scale, so recalibrating confidence cannot help. A monotone-nonlinear reshaping (isotonic regression of the posterior to the labels) does change the shape, yet recovers only $+12\%$ of $I(\hat{g})$ ($0.78 \rightarrow 0.87$). The $4.5\times$ shortfall is close to irreducible for these features: the per-token signal does not separate faithful from hallucinated tokens as sharply as the marginal feature divergence suggests. Shrinking it needs more discriminative features, not a retuned model.

The residual factor of two is finite-horizon, not an asymptotic correlation effect. The learned score is strongly smoothed in time: its clean-stream autocorrelation decays slowly ($\rho_1 = 0.94$), with integrated autocorrelation time $\tau \approx 22$. Correlated increments should, asymptotically, deflate the rate, and indeed the adjustment coefficient read from the detector’s own ARL_0 -threshold curve is $\omega^* \approx 0.044$, below the marginal $\omega = 0.95$ by precisely the factor τ . Yet the asymptotic dependent-data rate [6] then predicts a delay near 126 tokens, an order of magnitude past what we observe. It overshoots because detection here is *faster than mixing*: the onset is caught in ~ 11 tokens, well inside the score’s $\tau \approx 22$ correlation time, and the reset CUSUM floors ARL_0 at the mean document length. Neither limit is exact in this regime. The i.i.d. rate is a lower bound on delay (5.9); the realized 11.5 sits a factor of 2 above it, and pinning that down is a finite-horizon question, not an asymptotic correlation correction.

Low-false-alarm onset detection is intrinsically hard. Recall is near 30% at $\text{ARL}_0 = 100$ for every drift detector, so the recall-honest censored EDD is 56–66 tokens: most onsets are simply not caught within a tight false-alarm budget. This is consistent with Snel et al. [18], who find the first hallucinated token detectable at an AUC around 0.8 rather than 1.0. The first token of a span is the most detectable one, but it is far from trivially detectable, and a streaming monitor that must hold its false-alarm rate down will miss most onsets at their first token.

4.4 Robustness of the deficit to richer features and a rate-aware objective

The two prescriptions of Section 4.3 — more discriminative features, a better-shaped score — are testable, and we test both. They fail, and the way they fail says what the deficit is.

First the anatomy of the rate. For a centered reference the increment has $\mathbb{E}_0[Y] = -m$ and $\mathbb{E}_1[Y] = +m$ with $m = \frac{1}{2}(\mu_1 - \mu_0)$ the half mean-gap of the log-odds, and for near-Gaussian increments the adjustment coefficient is $\omega = 2m/\sigma_0^2$, so the realized rate is

$$I(\hat{g}) = \omega \delta_1 = \frac{2m^2}{\sigma_0^2}, \quad (4)$$

where σ_0 is the standard deviation of the score increment on clean tokens. The form holds on our data: the empirical I is within 10–25% of (4) across LogReg, ForwardGRU and BiGRU, the increments being close to Gaussian (Appendix F). Equation (4) also factors the deficit: $D/I = (D/D_{\text{inc}})(D_{\text{inc}}/I)$, where D_{inc} is the divergence of the scalar increment itself. The accumulation factor $D_{\text{inc}}/I \approx 1.1$ — the CUSUM is already near-optimal given the score — and the whole shortfall is $D/D_{\text{inc}} \approx 4$: the scalar log-odds is a lossy summary of the 33-dimensional feature law.

More features do not help, even when they carry genuine new divergence. The natural black-box candidate is self-consistency: resample the generator $K = 5$ times and score each token by how much the resamples contradict its sentence [9]. On RAGTruth this adds 0.83 nats of real, non-Gaussian divergence to the feature law (a nearest-neighbour estimate, against a random-feature control that adds none). Yet a ForwardGRU-CUSUM trained on the augmented features matches the baseline within seed noise on every axis we care about — token AUC, $I(\hat{g})$, and censored EDD (five seeds, Appendix F). The divergence is in the features; the trained scalar score does not realize it.

Retuning the objective does not help either, and (4) says why. The rate depends only on the standardized separation m/σ_0 , which is invariant to the scale of the log-odds. Adding a penalty on the clean-token increment variance σ_0 to the cross-entropy loss lowers σ_0 as intended, but lowers m in step: across penalty strengths m/σ_0 holds at 0.57 ± 0.01 and the censored EDD does not move ($55.4 \rightarrow 54.9$ tokens, within seed noise). This is the training-time twin of the temperature-scaling invariance in Section 4.3: a scale-only intervention, post-hoc or in the loss, cannot change a rate that is set by the score’s shape.

Both routes fail for one reason. The scalar score sits at the ceiling of what a one-dimensional causal statistic of these features can carry: m/σ_0 does not rise when the feature divergence is raised (the score does not pick it up) or when the objective is retuned (it only rescales). What the factorization isolates is the D/D_{inc} bottleneck — the compression of a high-dimensional feature law into one log-odds.

A multivariate accumulator of the mean shift is the natural way to lift that bottleneck, and on these signals it does not. Measured by delay among detected, an optimal linear readout of the recurrent hidden state (the LDA projection $\Sigma_h^{-1}(\mu_1 - \mu_0)$ accumulated as a CUSUM) matches the trained scalar head to within a token (12.7 against 11.5). A Hotelling generalized-likelihood-ratio statistic on the same 64-dimensional state is worse (18.1), defeated by the noise of a quadratic form in many dimensions (Appendix F). The cross-entropy head already reads the representation near-optimally for detection.

The bottleneck is also narrower than $D = 3.5$ suggests. That divergence counts variance and shape differences between the two regimes, but a drift CUSUM exploits only the mean shift, and on these signals the mean shift is the smaller part: the raw-feature Mahalanobis separation is 0.40 nats against the 3.5 the diagonal-Gaussian divergence reports, so the regimes differ more in the spread of the features than in their location.¹ The recurrent model concentrates the usable mean shift into its representation (1.5 nats) and the scalar drift captures most of it. The deficit is a property of the information a causal drift carries, not of the rule that accumulates it; closing it needs features whose location, not just their variance, separates the regimes — internal-state probes among them, at the cost of black-box applicability.

5 Discussion

Treating hallucination onset as a change-point buys something the classification view lacks: a yardstick. Token AUC rewards getting the average token right and says nothing about reaction time, whereas the Lorden floor states, in the units a deployment cares about, how fast any detector could possibly be. Read against that floor, our measurements carry two lessons.

The first is that the temporal frame is the right one but is not where the empirical win comes from. A causal recurrent labeler is a learned CUSUM, and the learning matters: it detects two to three times faster than a parametric CUSUM or a linear per-token model at a matched false-alarm rate. What the learning buys, though, is mostly a better per-token score. A nonlinear per-token classifier with no sequence already closes most of the gap to the recurrent CUSUM, leaving the sequential accumulation as a real but secondary effect and the extra causal context within bootstrap noise. We read this as a clarification of temporal modeling rather than a demotion of it: the sequential-detection framework is what makes the delay measurable and gives the recurrent labeler its CUSUM interpretation, while the magnitude of the empirical speedup is set by score quality.

The second lesson concerns the order-of-magnitude gap that remains between the best detector and the floor, and it is not an argument for deeper models. The delay of any score-based detector is fixed by the information rate its score realizes, and our learned score realizes only about a fifth of the divergence the features carry, a deficit that is invariant to recalibration and barely moves under monotone reshaping. The bound is a property of the information in the features, so the way to halve the achievable delay is to extract features that separate the two regimes more sharply, not

¹A detector built on that second-order structure recovers part of what the drift score misses. A CUSUM on the covariance term $\frac{1}{2}x^\top(\Sigma_0^{-1} - \Sigma_1^{-1})x$ of the full Gaussian log-likelihood ratio reaches a recall 0.065 higher than the best per-token scalar (a gradient-boosted classifier) at $\text{ARL}_0 = 100$ (95% bootstrap CI [0.03, 0.10]), at comparable per-detection delay and only on the open-weight generators. The mean term alone stays below the scalar, so the gain is the covariance change, not a location shift. The effect is modest and we do not pursue it here.

to add depth. This is a general statement about score-based onset detection, not a quirk of our architecture.

A third point is easy to miss behind an AUC and worth stating plainly. At the floor’s operating point our detectors catch under a third of onsets at their first token. Low-false-alarm streaming detection is genuinely hard: a monitor that promises to flag hallucinations as they happen, without burying the user in false alarms, will miss most onsets at the first opportunity and catch them a span later, if at all. That is the regime real deployments live in, and a token-level AUC hides it. Because the detector needs no access to the generating model, it remains applicable to closed-source APIs where white-box probes do not; the sequential view simply makes its true operating cost visible.

6 Conclusion

Hallucination onset is a change-point, and treating it as one buys a yardstick the classification view lacks: a hard floor on how fast any detector can react. For RAGTruth that floor is about 1.3 tokens at a 1% false-alarm rate, and it rests on a fact worth stating plainly, that the faithful/hallucinated state is a first-order Markov chain and nothing more elaborate is needed. Against that floor, a causal recurrent labeler acts as a learned CUSUM and detects, among the onsets it catches, two to three times faster than a parametric CUSUM or a linear per-token baseline, though a controlled decomposition shows most of that advantage is a better per-token score rather than the sequence, and the detector still sits an order of magnitude above the bound.

Two directions follow, and the analysis singles out the first as dominant. The larger lever is the divergence the score can realize: a feature set that doubles $D(P_1||P_0)$ halves the achievable delay, and unlike recalibration it actually moves the rate. The second is theoretical. The factor-of-two residual lives in a regime where detection is faster than the score mixes, so neither the i.i.d. first-order rate nor the asymptotic correlation correction is tight; a finite-horizon analysis of CUSUM delay under a strongly autocorrelated score would close it. Both are concrete, and both are measurable against the bound this paper puts on the table.

7 Limitations

The bound and the theorem rest on idealizations worth stating plainly, because each one bears on how the numbers should be read.

The first is that the emissions are not i.i.d. given the state. Theorem 1 and the divergence in (1) treat the per-token features as independent draws from P_0 or P_1 conditional on the state, yet several of our 33 features are windowed or cumulative, so consecutive tokens are correlated even within one regime. Correlated observations carry less information per token than i.i.d. ones, so the true per-token evidence is below $D(P_1||P_0)$ and the floor of 1.3 tokens is, if anything, optimistic; the gap we report is therefore a lower bound on the true gap.

The divergence itself is only a diagonal-Gaussian estimate. We compute $D(P_1||P_0) \approx 3.5$ nats under a diagonal-Gaussian model of the feature law, which ignores cross-feature dependence and non-Gaussian shape, and a different estimator would move the floor. We use this model because the naive CUSUM baseline assumes it too, which keeps that comparison fair, but the precise value of the bound should be read as an order of magnitude rather than a constant to three digits. As a check, a non-parametric k -nearest-neighbour estimator [19] of the same divergence gives $D \approx 2.8$ nats (floor ≈ 1.6 tokens), in line with the diagonal-Gaussian value, so the floor is robust to the estimator.

The delay rate is first-order, and its residual is measured rather than bounded. Proposition 1 expresses the delay through the realized information rate $I(\hat{g})$, but it is first-order in the threshold and treats the score increments as i.i.d. Section 4.3 shows the leftover factor of two is finite-horizon: the asymptotic correlation correction overshoots by an order of magnitude because detection precedes the score’s mixing time, so neither limit is tight. We measure that residual rather than bound it; a finite-horizon rate for CUSUM under a strongly autocorrelated score would be stronger, and harder.

Our analysis also covers only the first onset of each generation. The optimality in Corollary 1(i) is for a single change-point, so generations with several spans, and the question of re-arming the detector after a span ends, are out of scope; the first-passage view matches a deployment that stops at the first alarm rather than one that monitors continuously through a whole generation.

Finally, the numbers come from one corpus and one operating regime. The transition probabilities p, q , the divergence, and hence the bound are estimated on RAGTruth, so while the qualitative ordering of detectors should transfer, the specific values, and the ARL_0 at which recall collapses, are corpus-specific. The delay numbers come from a trained model (seed 42); we quantify document-level uncertainty with 95% bootstrap confidence intervals (Figure 1) and additionally seed-average the decomposition over five seeds (Appendix C), where the accumulation step stays robust (3.4 ± 0.7 tokens) and the causal-context step stays within noise, confirming the single-seed reading.

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Appendix

A Data and Label Estimation

We use RAGTruth [11] with its official train/test split. The test split has 2,700 generations (943 with at least one hallucinated span, 1,757 clean) and about 341,000 tokens at a 4.16% hallucination rate; the training split, on which the transition probabilities and the feature divergence are estimated, holds about 1.98 million tokens. A token is labeled 1 when it falls inside a human-annotated hallucination span and 0 otherwise.

The transition matrix is the maximum-likelihood estimate of a two-state chain. Every adjacent label pair (y_{t-1}, y_t) over the training generations is counted into a 2×2 table C , and each row is normalized:

$$p = \frac{C_{01}}{C_{00} + C_{01}}, \quad q = \frac{C_{11}}{C_{10} + C_{11}}.$$

This gives the onset hazard $p \approx 0.0044$ and the persistence $q \approx 0.907$, a ratio $q/p > 200$. The higher-order rows of Table 1 use the same count-and-normalize estimator with a length- k context. The labels carried by the saved detector posteriors match the reference RAGTruth labels up to a relabeling difference of about 0.05% of tokens, too small to move p , q , or the bound.

B Operating-Characteristic Estimation

We estimate ARL_0 on the concatenation of all hallucination-free generations. The detector is run over this clean stream at a fixed threshold, and ARL_0 is the total number of clean tokens divided by the number of distinct alarms, where an alarm is counted at each upward threshold crossing and the statistic is reset afterward. Because the stream is not segmented by document, ARL_0 is not capped by document length, unlike a per-document false-alarm rate. Delay is measured on the hallucination generations: the alarm position minus the onset θ when the detector fires at or after θ , and the maximum possible delay (tokens remaining after θ) when it misses or fires early, the latter giving the censored EDD. Detectors are matched by sweeping their thresholds to a common ARL_0 before any delay is compared.

C Speedup Decomposition and Bootstrap

The decomposition of Figure 1 fixes each detector’s threshold once on the full data at the $ARL_0 = 100$ operating point, then resamples the hallucination generations with replacement ($B = 1,000$) and recomputes the delay among detected at those fixed thresholds. This conditional bootstrap captures document-level variance without re-matching ARL_0 on every resample. The three steps are the nonlinear score (LogReg to HistGBM, -12.9 tokens, 95% confidence interval $[8.8, 17.0]$), the accumulation (HistGBM threshold to CUSUM, -4.5 , $[1.8, 7.1]$), and the causal context (HistGBM-CUSUM to ForwardGRU-CUSUM, -1.9 , $[-1.0, 4.7]$); the first two intervals exclude zero and the third does not. A ForwardGRU trained on token-shuffled sequences (15.6 tokens) isolates the value of temporal order to the recurrent model itself.

The bootstrap above varies documents at a fixed model; we also seed-average the decomposition by retraining all detectors under five seeds (0, 1, 2, 7, 42). The two recurrent-model-dependent steps survive training variance: the accumulation reduces the delay by 3.4 ± 0.7 tokens (robust, more than one standard deviation from zero), the causal-context step is 0.9 ± 1.6 tokens (within one standard deviation of zero), and the ForwardGRU-CUSUM delay is 12.2 ± 1.4 tokens. The document-variance

picture therefore carries over to training variance: the sequential accumulation is a real effect and the extra causal context is not.

D Models and Features

All detectors share the same 33-dimensional per-token feature vector (text statistics, NLI entailment, and generator log-probabilities) from the companion multi-signal system. The ForwardGRU is a two-layer unidirectional GRU of hidden width 64 with a sigmoid output head, trained by binary cross-entropy with AdamW (weight decay 10^{-4}) under seed 42 and a 15% stratified validation split. The HistGBM is a gradient-boosted classifier (500 trees, learning rate 0.05, up to 63 leaves per tree) on the same features, and the linear baseline is a class-balanced logistic regression. The full training recipe and the feature extractor are released with the companion system.

E Closing the Gap: Details

The realized information rate of the learned score is $I(\hat{g}) = \omega \delta_1 = 0.78$ nats per token, with Lundberg exponent $\omega = 0.95$ and post-change drift $\delta_1 = 0.82$, against a feature divergence $D \approx 3.5$ nats, so $D/I(\hat{g}) = 4.5$. Temperature scaling $s \mapsto s/T$ leaves $I(\hat{g})$ exactly invariant ($\omega \mapsto \omega T$, $\delta_1 \mapsto \delta_1/T$); isotonic recalibration of the posterior to the labels changes the score’s shape but recovers only +12% ($0.78 \rightarrow 0.87$). On the clean stream the score is strongly autocorrelated ($\rho_1 = 0.94$, integrated autocorrelation time $\tau \approx 22$), and the adjustment coefficient read from the detector’s own ARL_0 -threshold curve, $\omega^* \approx 0.044$, falls below the marginal ω by the factor τ . The asymptotic dependent-data rate then predicts a delay near 126 tokens, an order of magnitude past the observed 11.5, because detection occurs well inside the score’s correlation time; this is the finite-horizon residual of Section 4.3.

F Robustness of the Deficit

Rate anatomy. For a centered reference the score increment is $Y = \text{logit } \hat{p} - k$ with $k = \frac{1}{2}(\mu_0 + \mu_1)$, so $\mathbb{E}_0[Y] = -m$ and $\mathbb{E}_1[Y] = +m$ with $m = \frac{1}{2}(\mu_1 - \mu_0)$. Modeling the clean increment as $Y \sim N(-m, \sigma_0^2)$, the Lundberg condition $\mathbb{E}_0[e^{\omega Y}] = \exp(-m\omega + \frac{1}{2}\sigma_0^2\omega^2) = 1$ has the nonzero root $\omega = 2m/\sigma_0^2$, and since $\mathbb{E}_1[Y] = m$,

$$I(\hat{g}) = \omega \mathbb{E}_1[Y] = \frac{2m^2}{\sigma_0^2},$$

which depends on the score only through the standardized separation m/σ_0 and is therefore invariant to its scale. Table 3 checks the form against the empirical rate computed from the saved increments: the agreement is within 10–25%, the increments being close to Gaussian (skewness $|\cdot| < 0.5$).

Feature augmentation (self-consistency). We resample the generator $K = 5$ times at temperature 0.8 and score every token by the mean SelfCheck-NLI contradiction of its sentence against the resamples [9], on the Llama-2-7B-chat subset of RAGTruth. A nearest-neighbour estimate [19] puts the conditional divergence this block adds at +0.83 nats over the 33-dimensional law, against −0.59 for a matched block of random features and ≈ 0 under a diagonal-Gaussian model: the gain is real and non-Gaussian. Retraining the ForwardGRU-CUSUM on the augmented features changes nothing within five-seed noise: token AUC $0.758 \pm .007 \rightarrow 0.756 \pm .006$, $I(\hat{g})$ $0.572 \pm .029 \rightarrow 0.533 \pm .029$,

Score	m	σ_0	m/σ_0	I_{emp}	$I_{\text{emp}}/I_{\text{gauss}}$
LogReg	0.38	0.93	0.41	0.35	1.04
ForwardGRU	0.82	1.25	0.66	0.78	0.90
BiGRU	1.35	2.13	0.63	1.00	1.25

Table 3: The closed form $I = 2m^2/\sigma_0^2$ against the empirical rate. More context raises the standardized separation m/σ_0 (LogReg to the recurrent models), which is what lowers the deficit.

λ	AUC	σ_0	m/σ_0	censored EDD	recall
0 (BCE)	0.785	1.12	0.564	55.4±1.9	0.34
0.05	0.775	0.76	0.565	55.0±1.6	0.34
0.10	0.772	0.59	0.579	54.9±2.1	0.34
0.20	0.763	0.38	0.584	59.3±11.2	0.29

Table 4: A clean-variance penalty lowers σ_0 but not the standardized separation m/σ_0 , so the censored EDD does not move; $\lambda = 0.2$ begins to collapse. The rate is scale-invariant, so a scale-only objective cannot help.

censored EDD $51.3\pm.6 \rightarrow 52.6\pm1.9$ tokens (this subset’s baseline differs from the main table; the comparison is to its own baseline).

Rate-aware objective. Adding a penalty $\lambda \cdot \text{Var}_0[\text{logit}]$ on the clean-token log-odds variance to the cross-entropy loss lowers σ_0 but lowers m in proportion, leaving m/σ_0 — and the delay — unmoved (Table 4, five seeds, epoch chosen by held-out censored EDD). A discriminant-ratio objective $(\mu_1 - \mu_0)^2 / \text{Var}_0$ instead collapses: it drives $\sigma_0 \rightarrow 0$ and inverts the score, so we report the variance-penalty form. The realized rate I is itself gameable this way, which is why selection and the headline metric are the measured EDD, not I .

Multivariate accumulation. We replace the scalar log-odds with a multivariate statistic, in feature space and on the recurrent hidden state. The fixed projection is LDA, $w = \Sigma^{-1}(\mu_1 - \mu_0)$ with a shrinkage pre-change covariance, accumulated as a CUSUM; the adaptive one is a window-limited generalized likelihood ratio, $\text{GLR}_t = \max_{t-w \leq s < t} \|Z_t - Z_s\|^2 / (2(t-s))$ on the whitened cumulative deviations $Z = \sum \Sigma^{-1/2}(x - \mu_0)$, a Hotelling statistic that adapts the change direction to the data ($w = 50$). Table 5 reports the delay at $\text{ARL}_0 = 100$. On the raw features both are poor, since the change there is barely a shift in the mean (Mahalanobis separation 0.40 nats, against the 3.5 the diagonal-Gaussian divergence assigns to variance and shape). On the 64-dimensional hidden state the mean shift is real (1.5 nats) but the optimal linear readout only matches the trained scalar head, and the GLR is worse: a quadratic form over 64 dimensions raises the alarm threshold needed to hold ARL_0 , costing recall.

G Proofs

Throughout, $Y_t = s(X_t) - k$ are the centered score increments, the CUSUM is $S_t = \max(0, S_{t-1} + Y_t)$, and $\tau = \inf\{t : S_t \geq h\}$ is its stopping time. Write $\delta_1 = \mathbb{E}_1[Y] > 0$ for the post-change drift and let $\omega > 0$ be the Lundberg exponent solving $\mathbb{E}_0[e^{\omega Y}] = 1$.

Space	Detector	censored EDD	recall
features	LDA-CUSUM	59.6	0.34
features	GLR-CUSUM	80.6	0.02
hidden	scalar logit-CUSUM	56.3	0.31
hidden	LDA-CUSUM	56.2	0.31
hidden	GLR-CUSUM	66.1	0.22

Table 5: Multivariate detectors at $\text{ARL}_0 = 100$. On the raw features the change is not a mean shift, so a linear or quadratic accumulator gains nothing; on the hidden state the optimal linear readout (LDA) only matches the scalar head and the Hotelling GLR is worse. Reading the representation multivariately does not lift the deficit.

Proposition 1 (first-order delay). *Delay.* After the change the increments have positive mean δ_1 , so S_t is a random walk with drift δ_1 reflected at 0. Write the reflected walk as $S_t = \sum_{i \leq t} Y_i + L_t$ with $L_t \geq 0$ the accumulated reflection at the barrier. Wald’s identity on the free sum gives $\mathbb{E}_1[S_\tau] = \delta_1 \mathbb{E}_1[\tau] + \mathbb{E}_1[L_\tau]$; under the post-change positive drift the walk rarely returns to 0, so $\mathbb{E}_1[L_\tau] = O(1)$ is negligible relative to h . Neglecting it and the overshoot $S_\tau - h$ (bounded in expectation under a mild integrability condition on Y), $\mathbb{E}_1[S_\tau] = h(1 + o(1))$, hence

$$\text{EDD} = \mathbb{E}_1[\tau] = \frac{h}{\delta_1} (1 + o(1)).$$

False alarm. Under P_0 the increments have negative mean, so the walk drifts down and a false alarm is a large deviation. Tilt by the Lundberg exponent, $d\tilde{P}_0 = e^{\omega Y} dP_0$; by $\mathbb{E}_0[e^{\omega Y}] = 1$ this is a probability law under which the walk has positive drift, and the standard renewal estimate for the level-crossing of a tilted walk [17] gives $\text{ARL}_0 = \mathbb{E}_\infty[\tau] = e^{\omega h} (1 + o(1))$, i.e. $h = \omega^{-1} \ln \text{ARL}_0 (1 + o(1))$.
Combining,

$$\text{EDD} \approx \frac{h}{\delta_1} = \frac{\ln \text{ARL}_0}{\omega \delta_1} = \frac{\ln \text{ARL}_0}{I(s)}, \quad I(s) = \omega \delta_1. \quad \square$$

Theorem 2 (information-rate optimality). Write $D = D(p_1 \| p_0)$. The Donsker–Varadhan variational formula states that for every measurable f with $\mathbb{E}_0[e^f] < \infty$,

$$\mathbb{E}_1[f] - \log \mathbb{E}_0[e^f] \leq D, \quad (5)$$

with equality if and only if $e^f \propto dP_1/dP_0$ (P_0 -a.e.). Apply (5) with $f = \omega Y$. The Lundberg condition $\mathbb{E}_0[e^{\omega Y}] = 1$ makes the second term vanish, so

$$I(s) = \omega \mathbb{E}_1[Y] = \mathbb{E}_1[\omega Y] - \log \mathbb{E}_0[e^{\omega Y}] \leq D.$$

Equality holds iff $e^{\omega Y} \propto dP_1/dP_0$, i.e. $\omega Y = \log(p_1/p_0) + c$ for a constant c , i.e. s is an affine function of the log-likelihood ratio. For the delay statement, Proposition 1 gives $\text{EDD} \approx \ln(\text{ARL}_0)/I(s) \geq \ln(\text{ARL}_0)/D$, with the lower bound attained by the log-likelihood-ratio score, which is exactly the floor of Theorem 1. $\square$

Remark. The two results compose into the gap identity used in Section 4: at a false-alarm budget ARL_0 , the delay of a score s is $\ln(\text{ARL}_0)/I(s)$ and the floor is $\ln(\text{ARL}_0)/D$, so the multiplicative gap is $D/I(s) \geq 1$, the Donsker–Varadhan deficit of the score. The empirical $D/I(\hat{g}) \approx 4.5$ for the learned score is this deficit, measured.